

## 2015 GCSE Syllabus Worksheets

### Foundation Worksheets

Grades 1 to 5

Grade 1

Grade 2

Grade 3

Grade 4

Grade 5

Part 2

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#### Instructions for use

Clicking on something like this **Grade 4** will take you to the Grade 4 clip numbers and titles.



Clicking on something like this Compound Units. . . . . 142 will take you to Worksheet 142.



Clicking on something like this 

|                   |          |
|-------------------|----------|
| ©MathsWatch       | Clip 142 |
| Grade 4 questions |          |

 will take you back to the contents page.





Take me to . . .

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## Grade 3 Foundation and Higher

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Grade 3 continued 



1) Expand these brackets

a)  $2(x + 3)$

b)  $3(2x + 4)$

c)  $5(3p - 2q)$

d)  $4(x^2 + 2y^2)$

e)  $6(r - r^2)$

2) Expand these brackets

a)  $x(x - 2)$

b)  $x(3x + 5)$

c)  $p(3p - 7q)$

d)  $y(y + 6y^2)$

e)  $x(r + r^2)$

3) Expand these brackets

a)  $2x(x - 5)$

b)  $4x(2x + 3)$

c)  $5p(4p - 2q)$

d)  $2y(3y + 4x^2)$

e)  $x(x + r^2)$

4) Expand these brackets

a)  $x(x^2 - 2)$

b)  $3x(2x^3 + 1)$

c)  $5p^2(4p - 2)$

d)  $2y^2(3y^3 + 4y)$

e)  $2xy(x + y^2)$



## Simple Factorisation

1) Factorise

- a)  $2x + 4$
- b)  $2y + 10$
- c)  $3x + 12$
- d)  $3x - 6$
- e)  $5x - 15$

2) Factorise

- a)  $p^2 + 7p$
- b)  $x^2 + 4x$
- c)  $y^2 - 2y$
- d)  $p^2 - 5p$
- e)  $x^2 + x$

3) Factorise

- a)  $2x^2 + 6x$
- b)  $2y^2 - 8y$
- c)  $5p^2 + 10p$
- d)  $7c^2 - 21c$
- e)  $6x^2 + 9x$

4) Factorise

- a)  $2x^2 - 4xy$
- b)  $2t^2 + 10tu$
- c)  $6x^2 - 8xy$
- d)  $3x^2y^2 + 9xy$

## Substitution

1)  $y = 5x$

- a) Work out the value of  $y$  when  $x = 3$
- b) Work out the value of  $y$  when  $x = -2$

2)  $y = 2x + 7$

- a) Work out the value of  $y$  when  $x = 4$
- b) Work out the value of  $y$  when  $x = -3$

3)  $y = 2x + 4t$

$x = 6$

$t = 1$

Work out the value of  $y$ .

4)  $y = 2a - 3b$

$a = 4$

$b = -2$

Work out the value of  $y$ .

5)  $v = 3a + 5b$

$a = 6$

$b = -3$

Work out the value of  $v$ .

6)  $y = x^2$

- a) Work out the value of  $y$  when  $x = 6$
- b) Work out the value of  $y$  when  $x = -4$

7)  $y = 2x^2$

- a) Work out the value of  $y$  when  $x = 5$
- b) Work out the value of  $y$  when  $x = -3$

8)  $y = 3x^2 + 2x$

- a) Work out the value of  $y$  when  $x = 2$
- b) Work out the value of  $y$  when  $x = -4$



9)  $v = u^2 + 5as$

$u = 6$

$a = 2.5$

$s = 9$

Work out the value of  $v$ .



10)  $y = p - 2qx^2$

$p = -10$

$q = 2$

$x = -5$

Work out the value of  $y$ .



11)  $v^2 = u^2 + 2as$

$u = 6$

$a = 2.5$

$s = 9$

Work out the value of  $v$ .



12)  $v^2 = u^2 + 2as$

$u = 3$

$a = 9.8$

$s = 12$

Work out the value of  $v$ .  
Give your answer correct to 1 decimal place



13)  $s = ut + 0.5at^2$

$a = 9.8$

$t = 5$

$u = 7$

Work out the value of  $s$ .

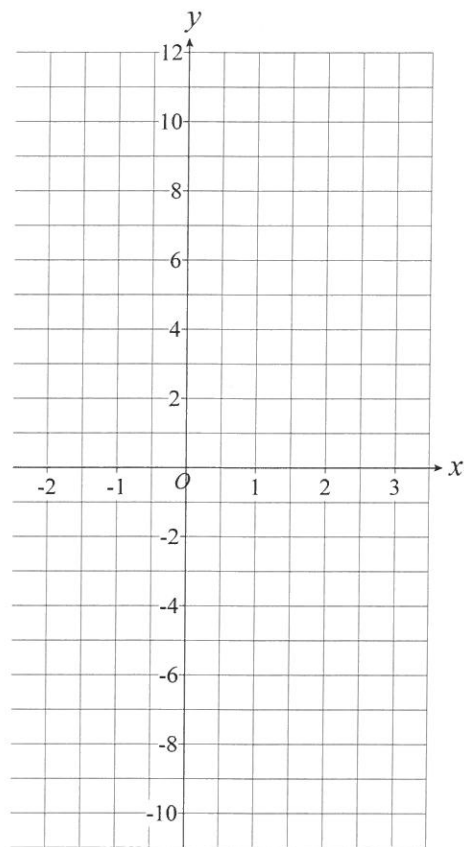


## Straight Line Graphs

- 1) a) Complete the table of values for  $y = 4x - 2$

|     |     |    |    |   |   |    |
|-----|-----|----|----|---|---|----|
| $x$ | -2  | -1 | 0  | 1 | 2 | 3  |
| $y$ | -10 |    | -2 |   |   | 10 |

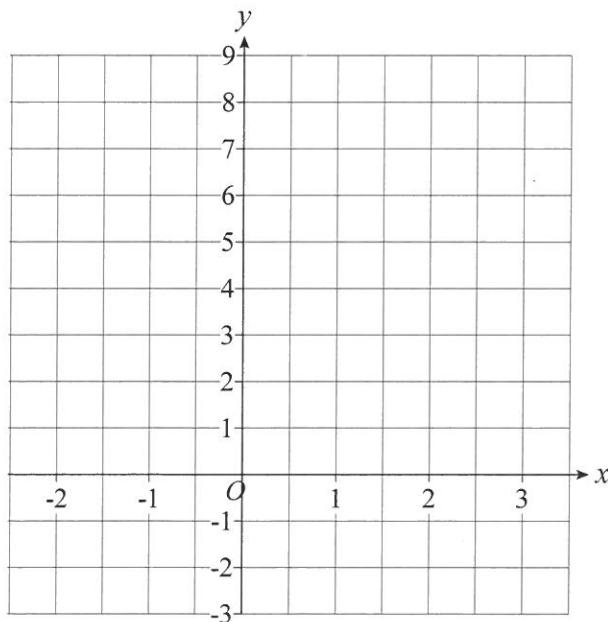
- b) On the grid, draw the graph of  $y = 4x - 2$ , for values of  $x$  from -2 to 3.
- c) Use the graph to find the value of  $y$  when  $x = 2.5$
- d) Use the graph to find the value of  $x$  when  $y = -8$



- 2) a) Complete the table of values for  $y = 2x + 2$

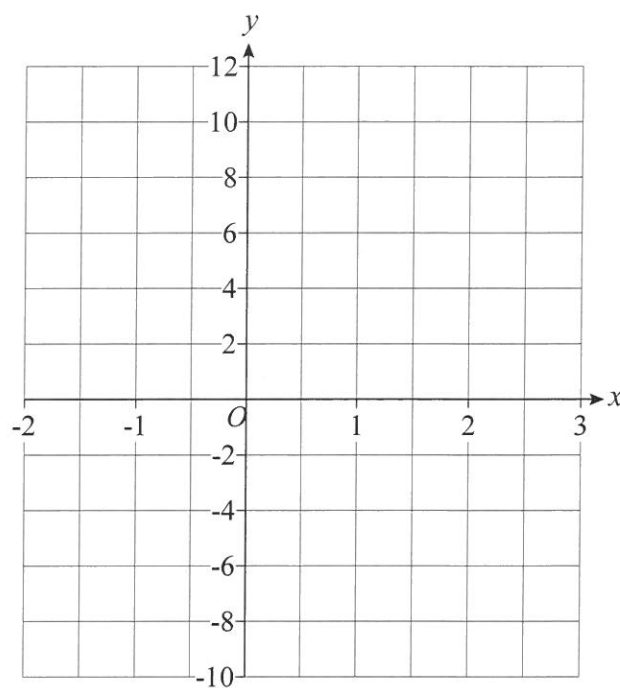
|     |    |    |   |   |   |   |
|-----|----|----|---|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    | 0  | 2 |   |   |   |

- b) On the grid, draw the graph of  $y = 2x + 2$ .



## Straight Line Graphs

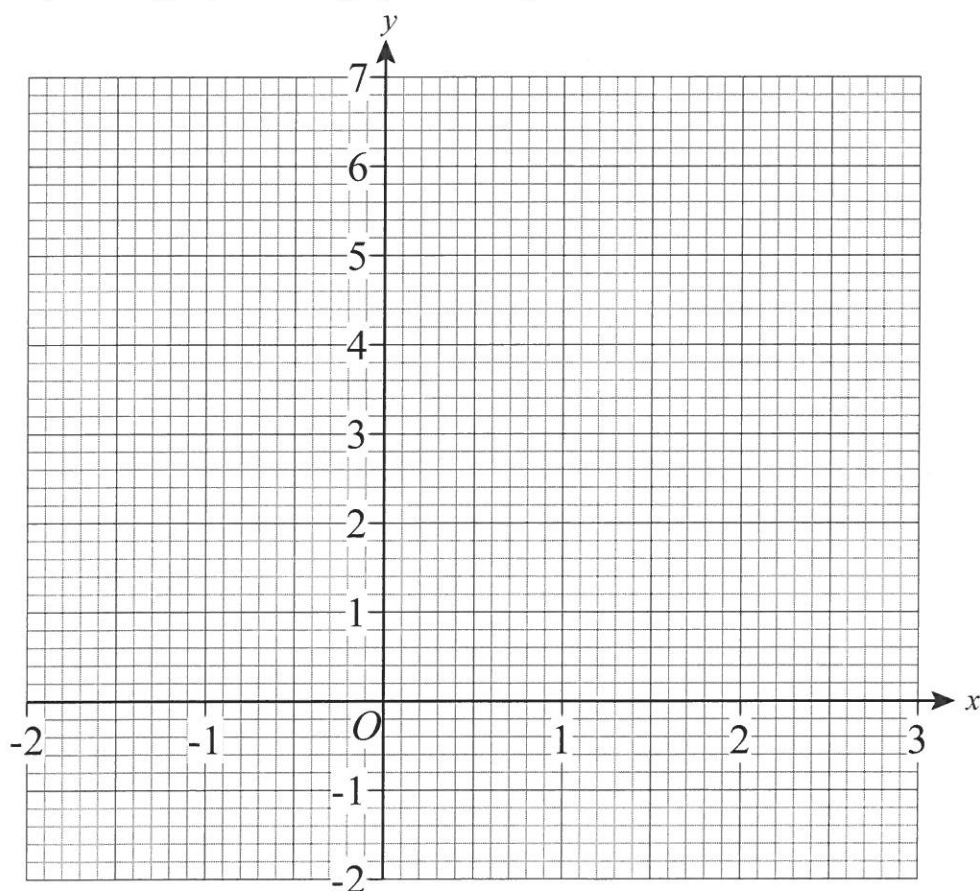
- 1) On the grid, draw the graph of  $y = 2x - 4$



- 2) a) Complete the table of values for  $3x + 2y = 6$

|     |    |     |   |   |   |      |
|-----|----|-----|---|---|---|------|
| $x$ | -2 | -1  | 0 | 1 | 2 | 3    |
| $y$ |    | 4.5 | 3 |   |   | -1.5 |

- b) On the grid, draw the graph of  $3x + 2y = 6$



- c) Find the gradient of the graph of  $3x + 2y = 6$ .

# Straight Line Graphs

- 1) a) Complete the table of values for  $y = 2x - 3$

|     |    |   |   |   |   |   |
|-----|----|---|---|---|---|---|
| $x$ | -1 | 0 | 1 | 2 | 3 | 4 |
| $y$ |    |   |   | 1 |   |   |

- b) Using the axes on the right draw the graph of  $y = 2x - 3$

- c) Use your graph to work out the value of  $y$  when  $x = 2.5$

- d) Use your graph to work out the value of  $x$  when  $y = 4.5$

\_\_\_\_\_

- 2) a) Complete the table of values for  $y = 2 - x$

|     |    |   |   |   |    |   |
|-----|----|---|---|---|----|---|
| $x$ | -1 | 0 | 1 | 2 | 3  | 4 |
| $y$ |    |   |   |   | -1 |   |

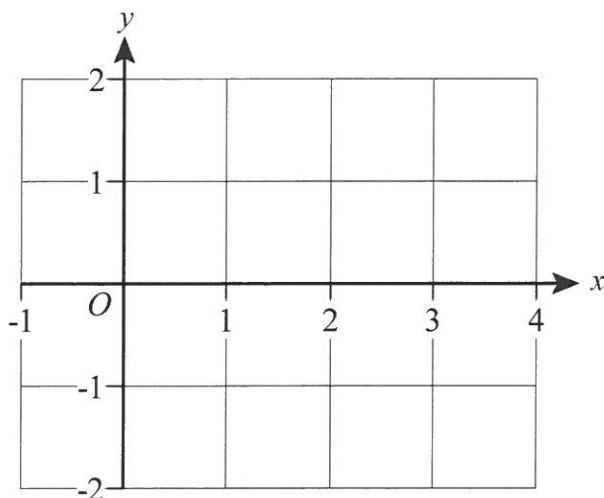
- b) Using the axes on the right, again, draw the graph of  $y = 2 - x$

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- 3) a) Complete the table of values for  $y = \frac{1}{2}x - 1$

- b) Draw the graph of  $y = \frac{1}{2}x - 1$

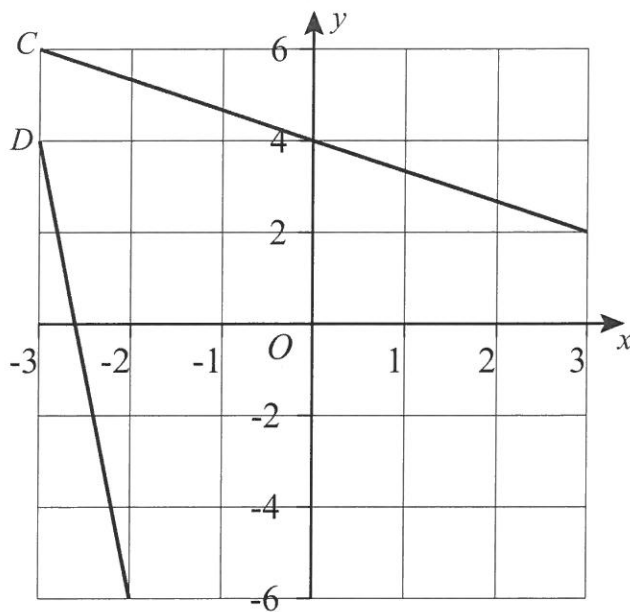
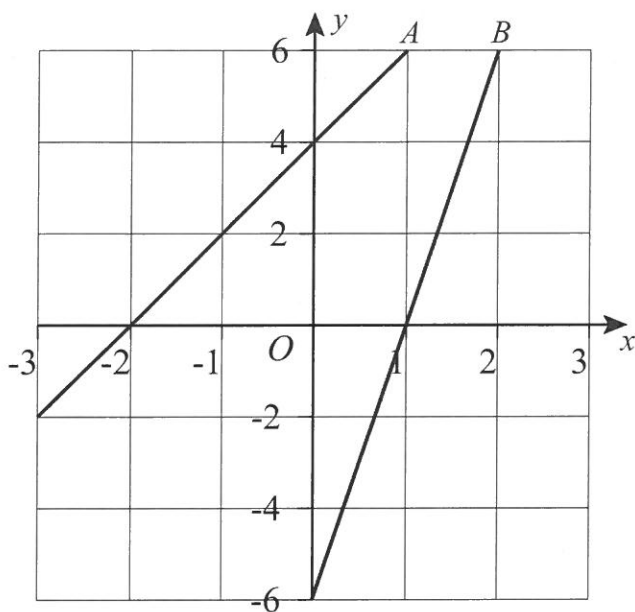
|     |    |   |   |   |   |   |
|-----|----|---|---|---|---|---|
| $x$ | -1 | 0 | 1 | 2 | 3 | 4 |
| $y$ |    |   |   | 0 |   |   |



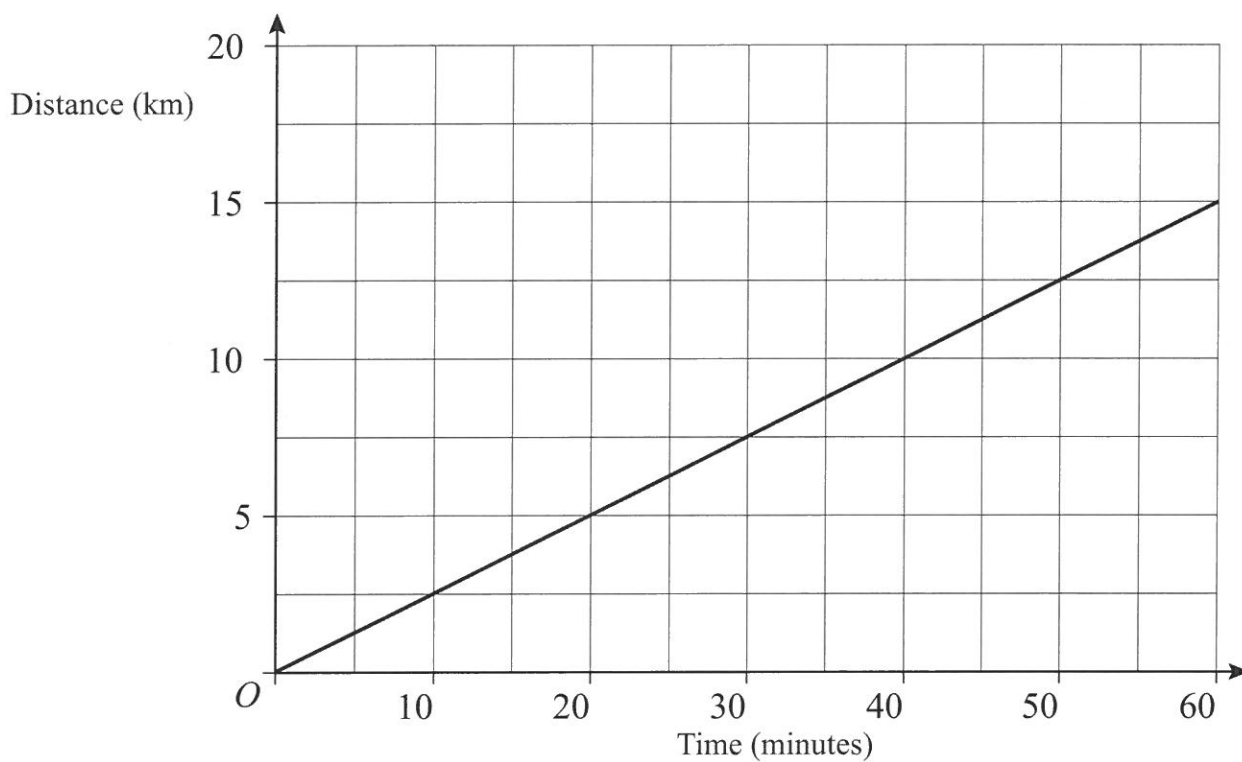
- c) Use your graph to find the value of  $y$  when  $x = 3.5$

## The Gradient of a Line

- 1) Find the gradient of lines  $A$ ,  $B$ ,  $C$  and  $D$ .



- 2) The graph shows how Meg cycles at a constant speed for 60 minutes.



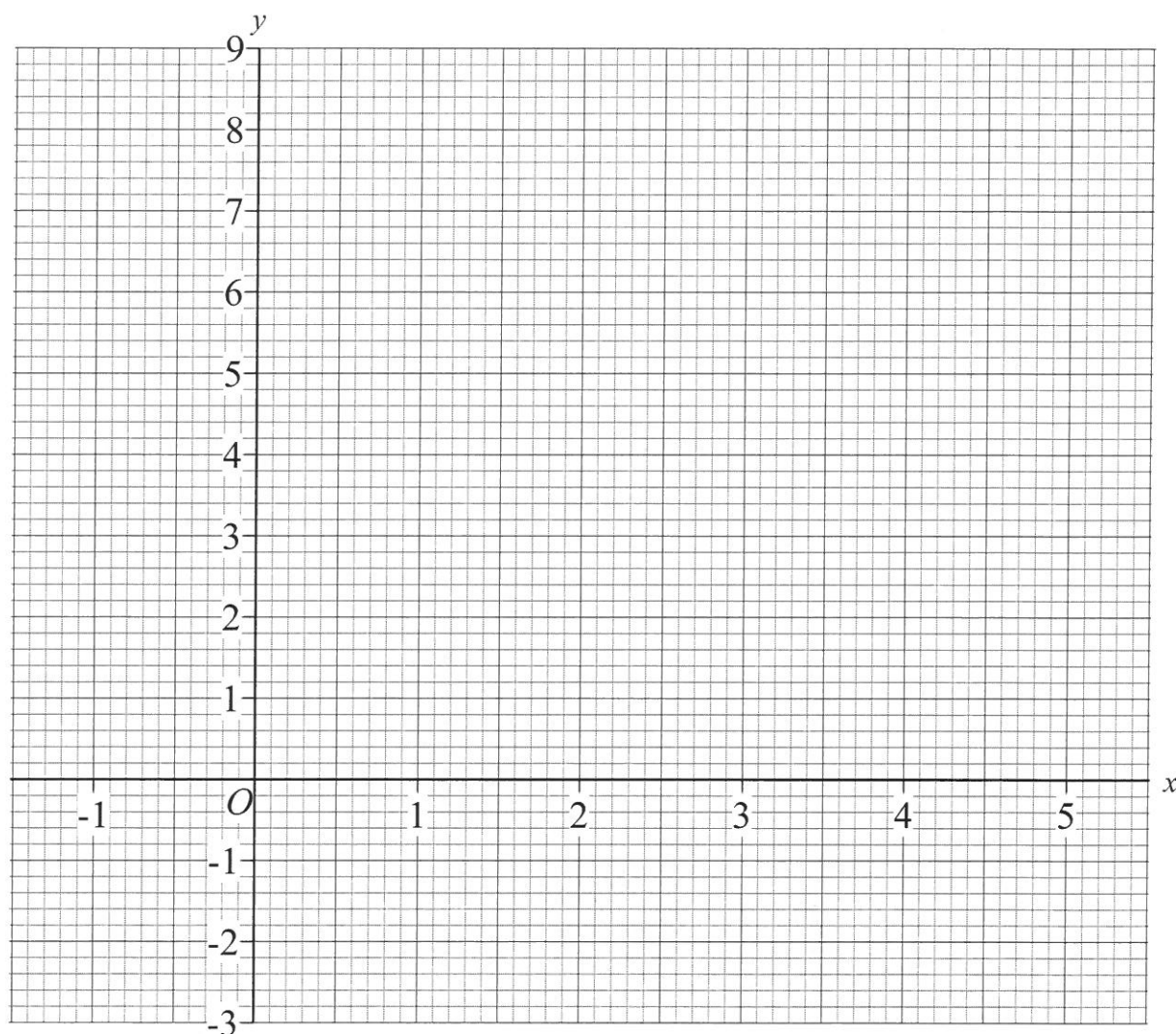
- Find the gradient of the line.
- What does the gradient show?

## Drawing Quadratic Graphs

1) Complete the table of values for  $y = x^2 - 4x + 3$

|   |    |   |   |   |   |   |   |
|---|----|---|---|---|---|---|---|
| x | -1 | 0 | 1 | 2 | 3 | 4 | 5 |
| y |    | 3 | 0 |   | 0 |   | 8 |

On the grid, draw the graph of  $y = x^2 - 4x + 3$

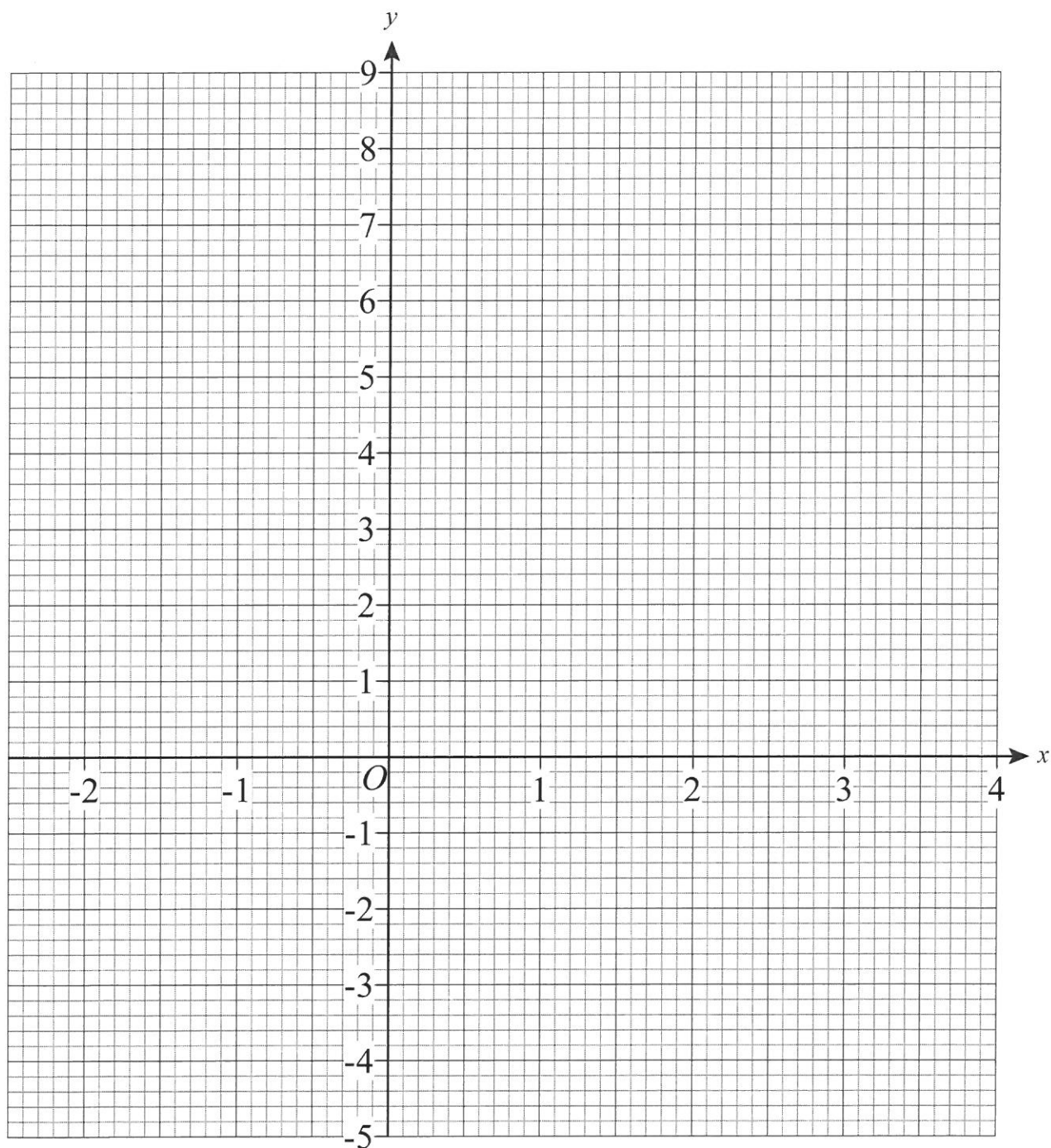


## Drawing Quadratic Graphs

1) a) Complete the table of values for  $y = x^2 - 3x - 2$

|   |    |    |    |    |   |    |   |
|---|----|----|----|----|---|----|---|
| x | -2 | -1 | 0  | 1  | 2 | 3  | 4 |
| y |    | 2  | -2 | -4 |   | -2 |   |

b) On the grid, draw the graph of  $y = x^2 - 3x - 2$



c) Use your graph to estimate the values of  $x$  when  $y = -1$

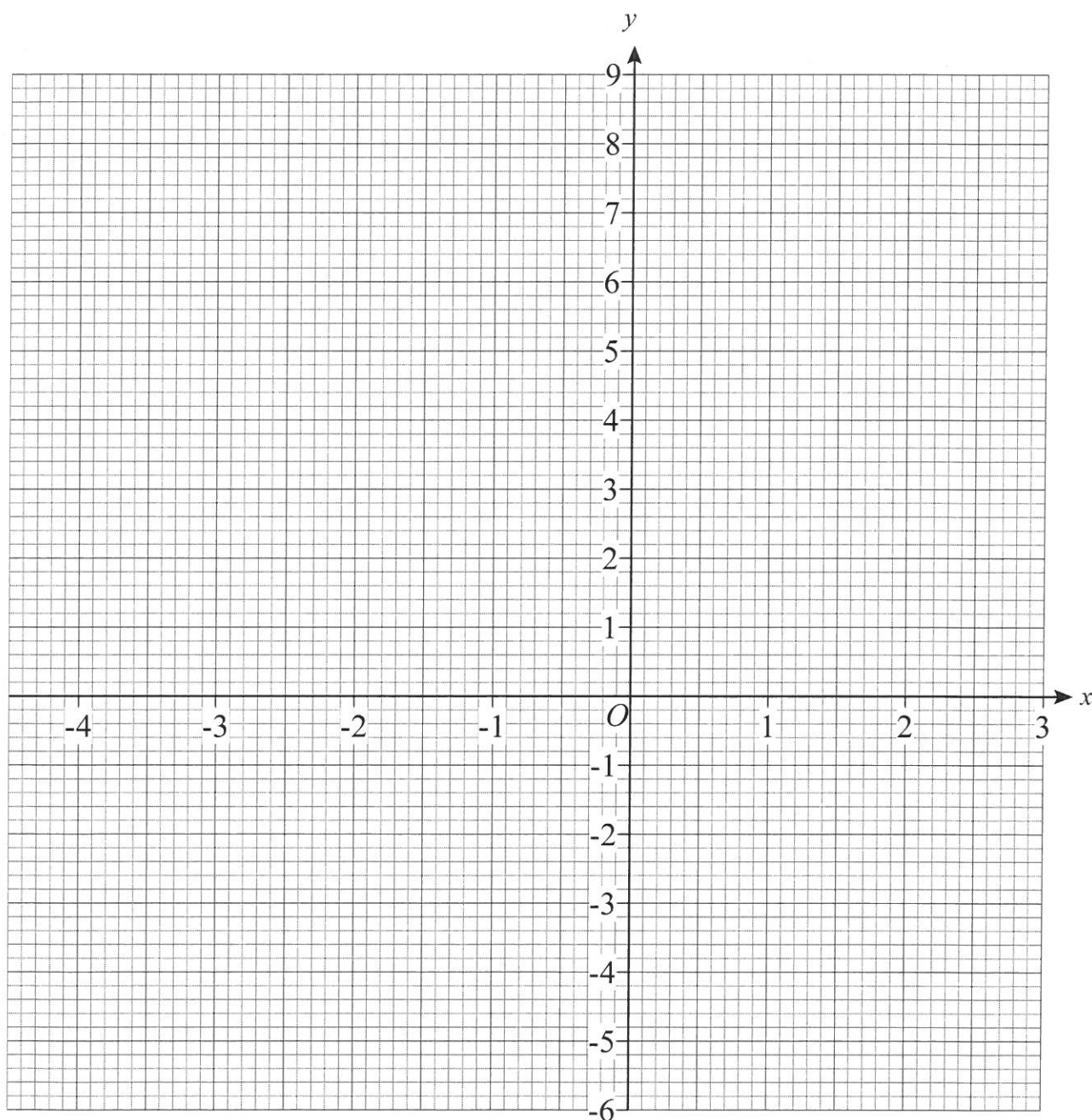
# Drawing Quadratic Graphs



1) a) Complete the table of values for  $y = x^2 + x - 4$

|   |    |    |    |    |   |   |   |   |
|---|----|----|----|----|---|---|---|---|
| x | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| y | 8  |    | -2 | -4 |   |   |   | 8 |

b) On the grid, draw the graph of  $y = x^2 + x - 4$





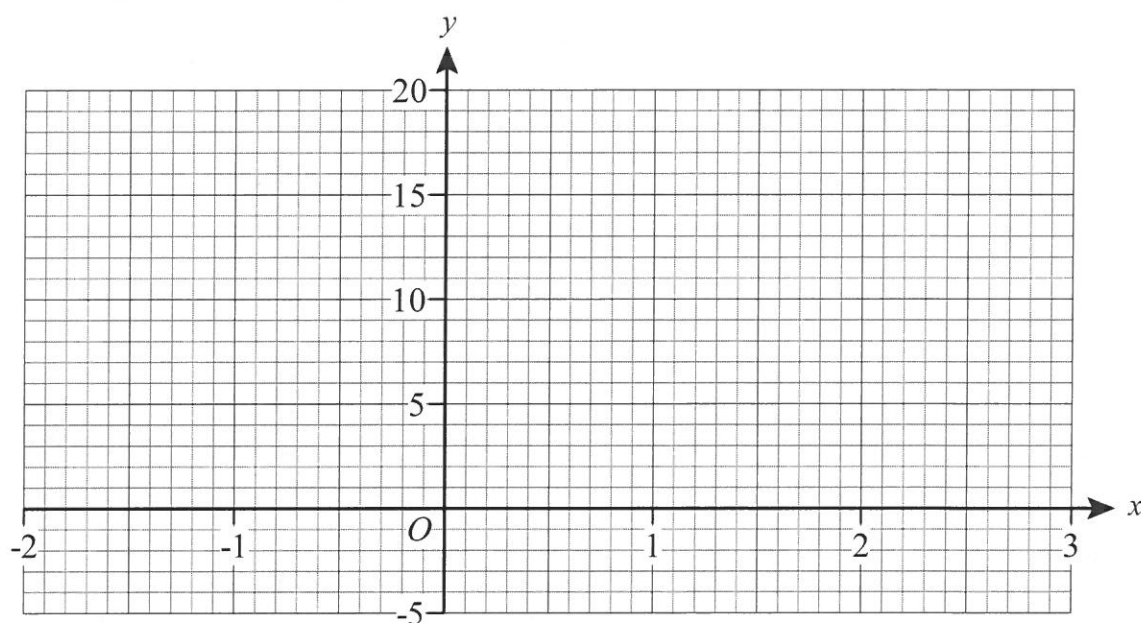
# Drawing Quadratic Graphs



- 1) a) Complete the table of values for  $y = 2x^2 - 3x$

|     |    |    |   |   |   |   |
|-----|----|----|---|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 14 |    | 0 |   |   | 9 |

- b) On the grid, draw the graph of  $y = 2x^2 - 3x$  for values of  $x$  from -2 to 3



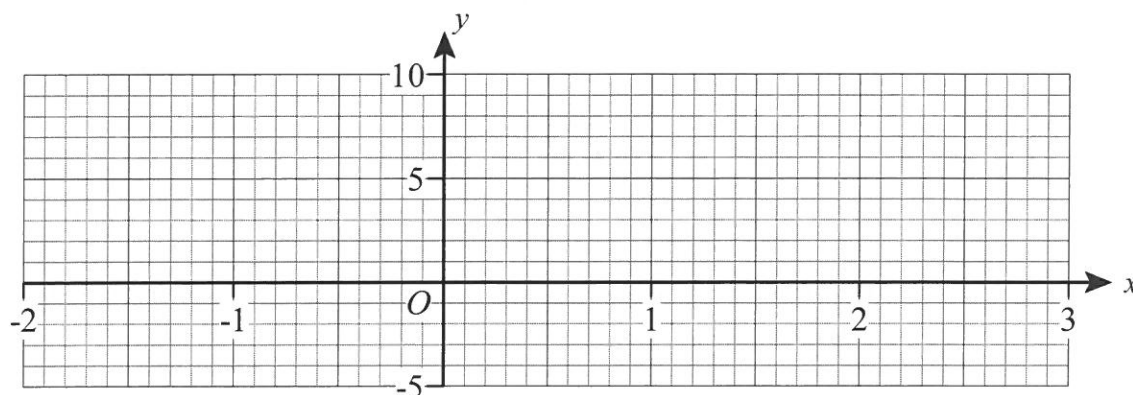
- c) Use the graph to find the value of  $y$  when  $x = -1.5$   
d) Use the graph to find the values of  $x$  when  $y = 4$



- 2) a) Complete the table of values for  $y = x^2 - 2x$

|     |    |    |   |   |   |   |
|-----|----|----|---|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 8  |    | 0 |   |   |   |

- b) On the grid, draw the graph of  $y = x^2 - 2x$  for values of  $x$  from -2 to 3

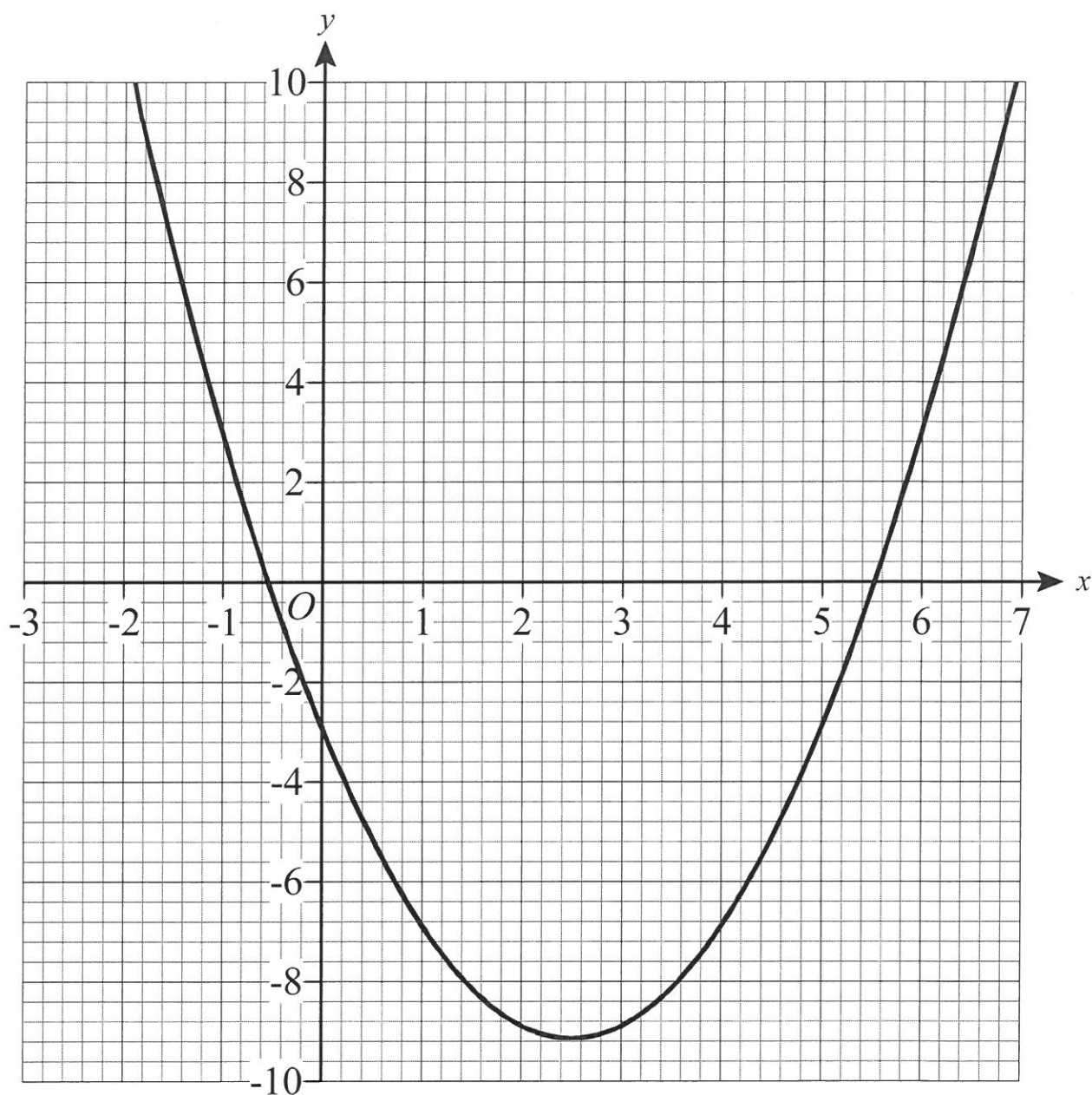


- c) (i) On the same axes draw the straight line  $y = 2.5$   
(ii) Write down the values of  $x$  for which  $x^2 - 2x = 2.5$



## Drawing Quadratic Graphs

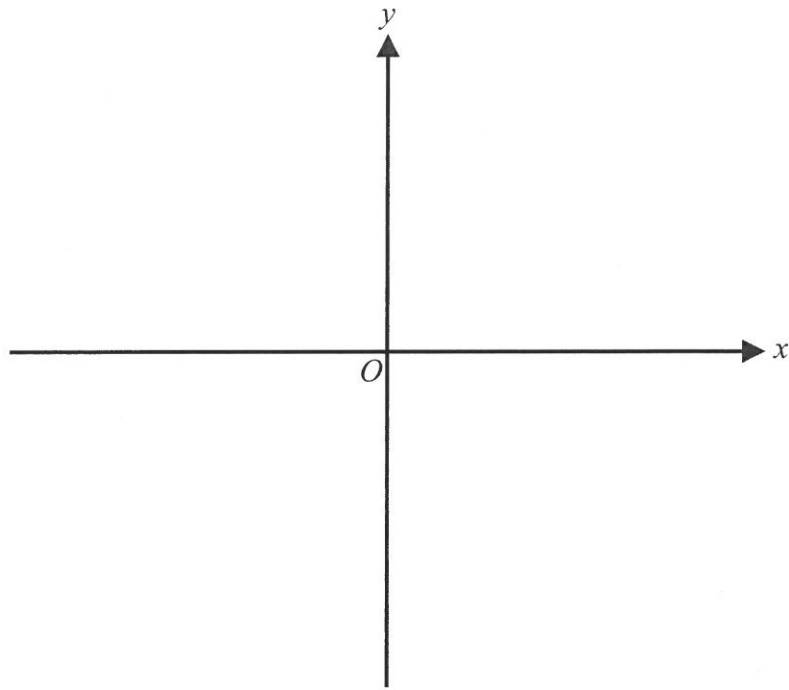
- 1) The diagram shows the graph of  $y = x^2 - 5x - 3$



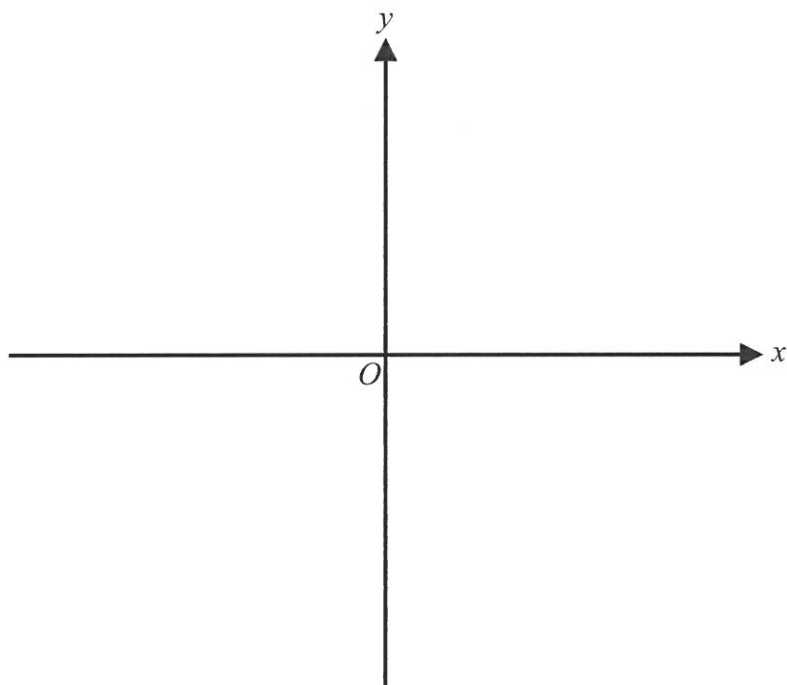
- a) Use the graph to find estimates for the solutions of
- $x^2 - 5x - 3 = 0$
  - $x^2 - 5x - 3 = 6$
- b) Use the graph to find estimates for the solutions of the simultaneous equations
- $$y = x^2 - 5x - 3$$
- $$y = x - 4$$

## Sketching Functions

- 1) a) Sketch the graph of  $y = 3x - 4$  on the axes, showing clearly where it crosses the  $y$ -axis.  
b) Sketch the graph of  $y = -2x + 3$  on the axes, showing clearly where it crosses the  $y$ -axis.



- 2) a) Sketch the graph of  $y = x^2 + 2$  on the axes, showing clearly where it crosses the  $y$ -axis.  
b) Sketch the graph of  $y = -x^2 - 1$  on the axes, showing clearly where it crosses the  $y$ -axis.



1) Solve the following equations.

a)  $2x - 7 = 11$

b)  $5x + 3 = 43$

2) Solve the following equations.

a)  $\frac{x}{5} + 1 = 7$

b)  $\frac{x}{2} - 6 = 2.5$

3) Solve the following equations.

a)  $2(4x - 1) = 46$

b)  $6(2x + 7) = 48$

4) Solve the following equations.

a)  $3\left(\frac{x}{7} + 2\right) = 6$

b)  $2\left(\frac{5x}{3} - 1\right) = 8$

- 1) Make  $x$  the subject of the formula  $w = 5x + 2$
  
- 2) Make  $x$  the subject of the formula  $y = 2x - 6$
  
- 3) Make  $x$  the subject of the formula  $2w = 3y + \frac{x}{5}$
  
- 4) Make  $t$  the subject of the formula  $a = 2(b + 3t) + 1$
  
- 5) Make  $x$  the subject of the formula  $y = 5 + \sqrt{x}$
  
- 6) Make  $t$  the subject of the formula  $w = x^2 + t$
  
- 7) Make  $n$  the subject of the formula  $m = 3n^2 - p$
  
- 8) Make  $q$  the subject of the formula  $2(5q^2 + 1) = c$

## Generate a Sequence from the $n$ th Term

- 1) The  $n$ th term of a number sequence is  $2n + 5$

Write down the first three terms of the sequence.

- 2) The  $n$ th term of a number sequence is  $3n - 1$

Write down the first four terms of the sequence.

- 3) The  $n$ th term of a number sequence is  $3n + 2$

Write down the first four terms of the sequence.

- 4) The  $n$ th term of a number sequence is  $5n - 7$

Write down the first four terms of the sequence.

- 5) The  $n$ th term of a number sequence is  $n^2$

Write down the first three terms of the sequence.

- 6) The  $n$ th term of a number sequence is  $n^2 + 3$

Write down the first three terms of the sequence.

- 7) The  $n$ th term of a number sequence is  $11 - n^2$

a) Find the third term of this sequence.

b) Find the fifth term of this sequence.

- 8) The  $n$ th term of a number sequence is  $n^2 + n$

a) Find the third term of this sequence.

b) Find the fifth term of this sequence.

- 1) Here are the first five terms of an arithmetic sequence.

1      3      5      7      9

Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.

- 2) Here are the first five terms of an arithmetic sequence.

6      10      14      18      22

Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.

- 3) Here are the first five terms of an arithmetic sequence.

1      4      7      10      13

Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.

- 4) Here are the first five terms of an arithmetic sequence.

7      12      17      22      27

Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.

- 5) Here are the first five terms of an arithmetic sequence.

8      6      4      2      0

Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.

- 1) Here are the first four terms of an arithmetic sequence.

4      7      10      13

Find an expression, in terms of  $n$ , for the  $n$ th term of the sequence.

- 2) The  $n$ th term of a number sequence is  $n^2 + 3$   
Write down the first three terms of the sequence.

- 3) Here are the first five terms of an arithmetic sequence.

2      7      12      17      22

- a) Find, in terms of  $n$ , an expression for the  $n$ th term of this sequence.
- b) An expression for the  $n$ th term of another sequence is  $11 - n^2$
- (i) Find the third term of this sequence.
- (ii) Find the fifth term of this sequence.
- 4) The  $n$ th term of a sequence is  $2n^2$
- (i) Find the 4th term of the sequence.
- (ii) Is the number 400 a term of the sequence?
- Give reasons for your answer.

- 1) The  $n$ th term of a number sequence is given by  $4n + 1$   
a) Work out the first **two** terms of the number sequence.

Here are the first four terms of another number sequence.

1   4   7   10

- b) Find, in terms of  $n$ , an expression for the  $n$ th term of this number sequence.

- 2) Here is a number pattern.

| Line Number |             |                    |    |
|-------------|-------------|--------------------|----|
| <b>1</b>    | $1^2 + 3^2$ | $2 \times 2^2 + 2$ | 10 |
| <b>2</b>    | $2^2 + 4^2$ | $2 \times 3^2 + 2$ | 20 |
| <b>3</b>    | $3^2 + 5^2$ | $2 \times 4^2 + 2$ | 34 |
| <b>4</b>    |             |                    |    |
| .           |             |                    |    |
| <b>10</b>   |             |                    |    |

- a) Complete Line Number 4 of the pattern.  
b) Complete Line Number 10 of the pattern.  
c) Use the number pattern to find the answer to  $999^2 + 1001^2$



- 1) What is the term to term rule for each sequence, below?
  - a) 1, 4, 16, 64, 256
  - b) 6, 12, 24, 48, 96
  - c) 80, 40, 20, 10, 5
  - d) 48, 12, 3, 0.75, 0.1875
  
- 2)
  - a) What is the special name for this sequence?  
1, 4, 9, 16, 25
  - b) What would be the 10th term of the sequence?
  - c) What is the  $n$ th term of the sequence?
  
- 3) By comparing the following sequences with the sequence of square numbers, find the  $n$ th term for each one.
  - a) 4, 7, 12, 19, 28
  - b) -1, 2, 7, 14, 23
  - c) 3, 12, 27, 48, 75
  
- 4)
  - a) What is the special name for this sequence?  
1, 3, 6, 10, 15
  - b) What would be the 8th term of the sequence?
  - c) What is the  $n$ th term of the sequence?
  - d) Use the  $n$ th term to work out the 21st term of the sequence.



- 1) Lance goes on holiday to France.  
The exchange rate is  $\text{£}1 = 1.15$  Euros.

He changes  $\text{£}350$  into Euros.

- a) How many Euros should he get?

In France, Lance buys a digital camera for 115 Euros.

- b) Work out the cost of the camera in pounds.



- 2) Whilst on holiday in Spain, Gemma bought a pair of sunglasses for 77 Euros.  
In England, an identical pair of sunglasses costs  $\text{£}59.99$ .

The exchange rate is  $\text{£}1 = 1.40$  Euros.

In which country were the glasses the cheapest, and by how much?

*Show all your working.*



- 3) Luke buys a pair of trainers in Switzerland.  
He can pay either 86 Swiss Francs or 56 Euros.

The exchange rates are:

$\text{£}1 = 2.10$  Swiss Francs

$\text{£}1 = 1.40$  Euros

Which currency should he choose to get the best price, and how much would he save?

*Give your answer in pounds (£).*



- 4) The exchange rate in London is  $\text{£}1 = \text{€}1.14$   
The exchange rate in Paris is  $\text{€}1 = \text{£}0.86$

Tony wants to change some pounds into euros.

In which of these cities would Tony get the most euros?

All working must be shown.

## Sharing Using Ratio

- 1) Tom and Julie share £48 in the ratio 5 : 3  
Work out how much more money Tom gets than Julie gets.
- 2) Ben and Sue share £60 in the ratio 2 : 3  
Work out how much each person gets.
- 3) A box contains milk chocolates and plain chocolates only.  
The number of milk chocolates to the number of plain chocolates is in the ratio 2 : 1  
There are 24 milk chocolates.  
Work out the total number of chocolates.
- 4) Andy, Ben and Claire share £54  
Ben gets three times as much money as Andy.  
Claire gets twice as much money as Ben.

How much money does Claire get?



- 5) There are some marbles in a bag.  
18 of the marbles are blue.  
12 of the marbles are red.
  - a) Write down the ratio of the number of blue marbles to the number of red marbles.  
Give your ratio in its simplest form.

There are some apples and pears in a box.  
The total number of apples and pears is 54.  
The ratio of the number of apples to the number of pears is 1 : 5

- b) Work out the number of pears in the box.

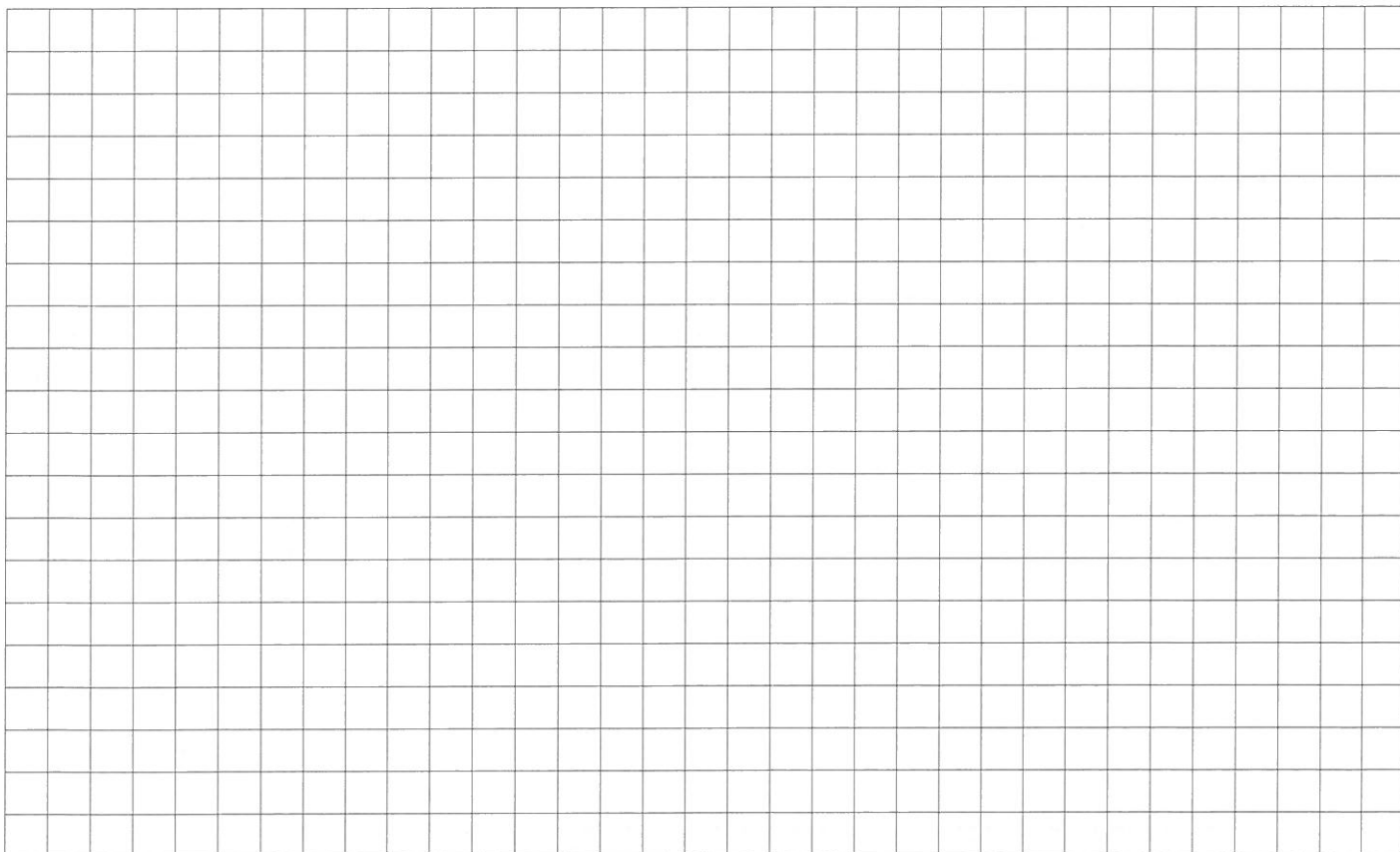


- 6) A piece of string is 180 cm long.  
Jim cuts it into three pieces in the ratio 2 : 3 : 4  
Work out the length of the longest piece.



- 7) Sally is 13 years old.  
Tammy is 12 years old.  
Danny is 10 years old.  
  
Sally, Tammy and Danny share £28 in the ratio of their ages.  
Tammy gives a third of her share to her mother.  
How much should Tammy now have?

- 1) In a box of chocolates, the ratio of plain chocolates to milk chocolates is 2 : 5.
  - a) What fraction of the chocolates are plain ones?
  - b) What fraction of the chocolates are milk ones?
  
- 2) If the ratio of  $x : y$  is 3 : 7, which of the following statements are correct?
  - a)  $x$  is  $\frac{3}{7}$  of  $(x + y)$
  - b)  $x$  is  $\frac{3}{10}$  of  $(x + y)$
  - c)  $y$  is  $\frac{7}{10}$  of  $x$
  - d)  $y$  is  $\frac{7}{10}$  of  $(x + y)$
  
- 3) Pounds can be converted to kilograms using the ratio 11 : 5.
  - a) Use the squares, below, to draw a conversion graph to illustrate this.
  - b) Convert 24 pounds to kilograms.
  - c) Convert 14 kilograms to pounds.



## Increase/Decrease by a Percentage

- 1) Increase:
 

|               |              |
|---------------|--------------|
| a) 500 by 10% | c) 80 by 15% |
| b) 320 by 10% | d) 75 by 20% |
  
- 2) Decrease:
 

|               |               |
|---------------|---------------|
| a) 400 by 10% | c) 140 by 15% |
| b) 380 by 10% | d) 35 by 20%  |
  
- 3) The price of a laptop is increased by 15%.  
The old price of the laptop was £300.  
Work out the new price.
  
- 4) The price of a £6800 car is reduced by 10%.  
What is the new price?



- 5) Increase:
 

|               |                 |
|---------------|-----------------|
| a) 65 by 12%  | c) 600 by 17.5% |
| b) 120 by 23% | d) 370 by 17.5% |



- 6) Decrease:
 

|              |                |
|--------------|----------------|
| a) 42 by 15% | c) 52 by 8.5%  |
| b) 79 by 12% | d) 8900 by 18% |



- 7) The price of a mobile phone is £78.40 plus VAT.  
VAT is charged at a rate of 17.5%.  
What is the total price of the mobile phone?



- 8) In a sale, normal prices are reduced by 7%.  
The normal price of a camera is £89.  
Work out the sale price of the camera.



- 9) A car dealer offers a discount of 20% off the normal price of a car, for cash.  
Peter intends to buy a car which usually costs £6800.  
He intends to pay by cash.  
Work out how much he will pay.



- 10) A month ago, John weighed 97.5 kg.  
He now weighs 4.5% more.  
Work out how much John now weighs.  
Give your answer to 1 decimal place.



- 1) A car dealer is comparing his sales over the past two years.  
In 2006, he sold 175 cars.  
In 2007, he sold 196 cars.

Work out the percentage increase in the number of cars sold.



- 2) In September 2005, the number of pupils attending MathsWatch College was 1352.  
In September 2006, the number of pupils attending MathsWatch College was 1014.

Work out the percentage decrease in the number of pupils attending MathsWatch College.



- 3) The usual price of a shirt is £32.50  
In a sale, the shirt is reduced to £29.25

What is the percentage reduction?



- 4) Olivia opened an account with £750 at the MathsWatch Bank.  
After one year, the bank paid her interest.  
She then had £795 in her account.

Work out, as a percentage, MathsWatch Bank's interest rate.



- 5) Keith buys a house for £270 000 and sells it two years later for £300 000.  
What is his percentage profit?  
Give your answer to 2 significant figures.



- 6) Shelley bought some items at a car boot sale and then sold them on ebay.  
Work out the percentage profit or loss she made on each of these items.
- a) Trainers bought for £15, sold for £20
  - b) DVD recorder bought for £42, sold for £60.90
  - c) Gold necklace bought for £90, sold for £78.30
  - d) A DVD collection bought for £120, sold for £81.60

## Reverse Percentage Problems



- 1) In a sale, normal prices are reduced by 20%.  
The sale price of a shirt is £26  
Calculate the normal price of the shirt.



- 2) A car dealer offers a discount of 15% off the normal price of a car for cash.  
Emma pays £6120 cash for a car.  
Calculate the normal price of the car.



- 3) In a sale, normal prices are reduced by 13%.  
The sale price of a DVD recorder is £108.75  
Calculate the normal price of the DVD recorder.



- 4) A salesman gets a basic wage of £160 per week plus a commission of 30% of the sales he makes that week.  
In one week his total wage was £640  
Work out the value of the sales he made that week.



- 5) Jason opened an account at MathsWatch Bank.  
MathsWatch Bank's interest rate was 4%.  
After one year, the bank paid him interest.  
The total amount in his account was then £1976  
Work out the amount with which Jason opened his account.



- 6) Jonathan's weekly pay this year is £960.  
This is 20% more than his weekly pay last year.  
Tess says "This means Jonathan's weekly pay last year was £768".  
Tess is wrong.  
a) Explain why  
b) Work out Jonathan's weekly pay last year.



- 7) The price of all rail season tickets to London increased by 4%.  
a) The price of a rail season ticket from Oxford to London increased by £122.40  
Work out the price before this increase.  
b) After the increase, the price of a rail season ticket from Newport to London was £2932.80  
Work out the price before this increase.

- 1) Meg has £1200 in her savings account.  
The account pays 5% simple interest per year.

How much interest will she earn in 4 years?

- 2) Dan has £4000 in his savings account.  
This account pays 2% interest per year.

How much interest will he earn in 6 years?

- 3) Chris borrows £6000 at a simple interest rate of 10% per year.  
He pays the money back after 4 years.

How much does he pay back in total?

- 4) Lisa borrows £3000 at a simple interest rate of 2.5% per year.  
She pays the money back after 3 years.

How much does she pay back in total?

- 5) Kate borrows £2000 at a simple interest rate of 16% per year.  
She pays the money back after 3 months.

How much does she pay back in total?

- 6) Neil invested £8000 in a savings account for 2 years.  
He earned £640 simple interest over the two years.

What was the interest rate?